

PERSONAL INFORMATION

Ahmed Mujtaba



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I am an ambitious, self-motivated, responsible, and hard-working person. I have experience working with artificial intelligence, embedded/edge devices, and computer vision. I worked in projects with Hyundai-Mobis and KIA, focusing on deep learning solutions for automotive industry. My experience includes hardware-software co-design, model deployment on edge devices, and model compression techniques. I conducted research on TinyML, published papers on neural network quantization and deployment on embedded devices. My research interests focus on EdgeAI, model compression, and TinyML. I am always excited to study new technological concepts to expand my knowledge. My career goal is to work in a diverse environment that offers a competitive work environment and innovative educational opportunities that will expand my skill set and result in useful inventions.

RESEARCH EXPERIENCE

Oct 2024 – present

Junior Scientist (PhD Student)

Silicon Austria Labs, GmbH, Austria

Website: <https://silicon-austria-labs.com/>

- Activities:
- Research on lightweight collaborative learning for edge devices.
 - Research on efficient machine learning models optimization techniques such as quantization, pruning, and knowledge distillation.
 - Contribute in communication activities to share research results with the general public and aside from their scientific community.
 - Conduct two-to-six months of secondment(s) at academic and intersectoral industrial partner organizations.
 - Publish in scientifically high-ranked conferences and journals for successful graduation.

Mar 2023 – Sept 2024

AI Researcher

DeltaX Co., Ltd, Seoul, South Korea

Website: <http://deltax.ai/>

- Activities:
- Collaborative project with Hyundai-Mobis, KIA, and many start-up companies in AI software.
 - End-to-End deep learning solutions for the automotive industry for applications, such as Driver Monitoring Systems (DMS), In-Cabin Monitoring Systems (ICMS), etc.
 - Hardware-Software co-designed DL/ML models and their deployment on edge devices such as Nvidia Jetson, Texas Instruments ADAS Chips (TDA4X), HAILO, etc.
 - Hands-on experience in embedded boards (such as Nvidia, Texas Instruments, HAILO, etc) and their SDKs for booting OS images on embedded devices.
 - Design end-to-end application deployment pipeline of computer vision solutions for their deployment on edge devices.
 - Experienced in deep learning model optimization methodologies (quantization, pruning etc).
 - Proficiency in designing efficient application pipelines for DL models using Python multi-threading and multiprocessing.
 - Research on model compression techniques such as quantization, pruning, knowledge distillation, etc.

Mar 2021 – Feb 2023

Graduate Research Assistant

ISML Lab, Gachon University, South Korea

Website: https://ai-security.github.io/professor_main_e.htm

- Activities:**
- Rich research background in TinyML and its applications.
 - Research publications and hands-on quantization and pruning of deep neural networks for image classification and object detection.
 - Research publications on deep learning optimization techniques for efficient deployment on resource-constrained environments such as microcontrollers.
 - Deployment of deep neural networks on ARM Cortex-M processors.
 - Hands-on experience with computer vision tasks such as image classification and object detection.

Oct 2019 – Nov 2019 **Internship** QONCEPT LABS, Islamabad, Pakistan

- Investigated latest advances in the field of Natural Language Processing (NLP) and Large Language Models (LLMs).
- Conducted a thorough survey for ongoing research in the LLMs.

EDUCATION

Oct 2024 – present **Doctoral Studies** University of Innsbruck, Austria

Thesis: "Collaborative Learning on the Edge".

Mar 2021 – Feb 2023 **Master of Science, Computer Engineering** Gachon University, Republic of Korea

Thesis: "TinyML: Efficient Quantization and Deployment of Deep Neural Networks for Embedded IoT Applications".

CGPA: 4.19/4.5 (93.25%)

Sept 2016 – Aug 2020 **Bachelor in Science, Electrical Engineering** UET, Taxila, Pakistan

Thesis: "Swarm Communication via Mobile Adhoc Networks (MANETs)"

CGPA: 3.53/4.0 (88.25%)

RESEARCH PUBLICATIONS

Journal Articles **A. Mujtaba**, W. -K. Lee and S. O. Hwang, "Low Latency Implementations of CNN for Resource-Constrained IoT Devices," in IEEE Transactions on Circuits and Systems II: Express Briefs, 2022, DOI: 10.1109/TCSII.2022.3205029.

Mujtaba, A., Lee, W.K., Ko, B.C. et al. AuGQ: Augmented quantization granularity to overcome accuracy degradation for sub-byte quantized deep neural networks. Appl Intell 55, 589 (2025). <https://doi.org/10.1007/s10489-025-06495-1>.

Conferences **Mujtaba, A.**, Radchenko, G., Prodan, R. and Masana, M., 2025. Federated Distillation on Edge Devices: Efficient Client-Side Filtering for Non-IID Data. arXiv preprint arXiv:2508.14769.

Rozi Khan, **Ahmed Mujtaba**, Wai Kong Lee and Seong Oun Hwang, Evaluation of Additive Power of Two Quantization on Various Deep Learning Datasets, The 9th International Conference on Green and Human Information Technology (ICGHIT), pp. 72-74, January 13-15, 2021.

Workshops **A. Mujtaba**, Majeed. A. and S. O. Hwang, "MLOps: Open Challenges from Hardware and Software Perspective in TinyML Devices" in Challenges in deploying and monitoring ML systems workshop - NIPS 2022

PATENTS

Han Kyung-hyun, **Mujtaba Ahmed**, Hwang Seong-un, Computing Device Performing Quantization of a Neural Network and Operating Method of the Computing Device, (Korean Patent Application No. 1020230032595), 2023. 03. 13.) <https://doi.org/10.8080/1020230032595>

SKILLS

Programming:	C/C++, Python, JavaScript, MATLAB, Verilog
AI Library/Framework:	TensorFlow, PyTorch
Hardware Platforms:	Arduino, STM32 Micro-controllers, Texas Instruments (e.g., TDA4X etc.), NVIDIA Jetson, HAILO-15 SBC
Embedded ML/DL:	CMSIS-NN, CMix-NN, STMCube.AI
Deployment Tools:	Docker
Collaborative Tools:	Git, GitHub, Confluence, JIRA
Operating System:	Linux, Windows
Soft Skills:	Project management, Strong decision-maker, Leadership, Complex problem solver, Innovative

PROJECTS

Mar 2021 – Feb 2023	FAST(Fast, Advanced, Secure and Transaction) Artificial Intelligence Convergence BK21 FOUR (Fostering Outstanding Universities for Research), Ministry of Education in Korea – Survey on the deployment on edge-based lightweight DL models. – Propose novel optimization techniques and perform experimentation to improve the inference latency of the DL models. – Publish novel research-based articles for premium journals.
May 2021 – May 2022	EQDL: Efficient Quantization Techniques for Deep Learning in Embedded System Korean IITP International Collaboration Funding, Republic of Korea Collaborator: University of Birmingham, Birmingham, England – Analyzed a suitable micro-controller platform to perform DL network inference and developed a baseline implementation of the proposed optimization techniques for efficient inference. – Optimized the software implementation of the baseline quantization libraries for microcontrollers (CMSIS/CMix) and utilized hardware resources in STM32H7. – Published novel research-based articles for premium journals.

EXTRACURRICULAR ACTIVITIES

Jul 2022 – Dec 2023	Chair Person IEEE Sensors Council Student Branch Chapter Gachon University, Seoul Section, Region 10 Organize and manage invited talks, training and mentoring activities, technical gatherings and networking programs.
Apr 2022 – Dec 2023	Member

IEEE Student Branch - Gachon University
Gachon University
Seoul Section, Region 10

Sep 2019 – Aug 2020 **Member**

IEEE
General Secretary of Computer Science Society
Served as a Member of IEEE CASE Student Branch, Islamabad Section, Region 10.

HONOURS AND AWARDS

February 2023 **Best Paper Award**

Gachon University
1342 Seongnam-daero, Bokjeong-dong, Sujeong-gu, Seongnam-si, Republic of Korea
Best paper award on publishing journal in IEEE Transaction of Circuits and Systems - II, Express Briefs, awarded by Gachon University, Republic of Korea

Mar 2021 – Feb 2023 **Global Scholarship**

Gachon University
1342 Seongnam-daero, Bokjeong-dong, Sujeong-gu, Seongnam-si, Republic of Korea
Awarded by Gachon University, Republic of Korea, for Master Degree in Computer Engineering

Sep 2016 – Aug 2020 **Distinction**

University of Engineering, Taxila
UET - HMC Link Rd, Taxila, Rawalpindi, Punjab 47050, Pakistan
Received Bachelor's degree with distinction in Electrical Engineering from University of Engineering (UET), Taxila, Pakistan.

Sep 2019 – Aug 2020 **Ignite Grant**

National Grassroot Information and Communication Technologies Research Initiative (NGIRI)
Ignite-National Technology Fund, Pakistan
Granted NGIRI funding from Ignite - National Technology Fund for Bachelor's Final Year Project: Swarm Communication via Mobile Adhoc Networks (MANETs)

LANGUAGE SKILLS

Mother tongue Urdu

Other languages

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken interaction	Spoken production	
English	C1	B2	C1	C1	B2

Levels: A1 and A2: Basic user – B1 and B2: Independent user – C1 and C2: Proficient user
[Common European Framework of Reference for Languages](#)